QC STORY PROJECT

PROJECT TITLE :-

To eliminate the concern of RHS Engine mounting insulator bolt not fit/ slip/ broken in Marazzo model.

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	15.06.19	25.06.19
Completion Date	30.10.19	30.10.19

Current Status:

closed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 3: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

1. Select the Theme

Brief description of Problem :-
To eliminate the concern of RHS Engine mounting insulator bolt not fit/ slip / broken in Marazzo model**.**

What :- Engine mounting insulator bolt not fit/ slip/ broken in Marazzo model.

Where :- Engine drop stage.

When :- From April -2019.

Why **:-** V1 concern , Major contributor line loss and in FRC loss, Rework is difficult.

Who :- Line operator.

How :- Avg. 44 concerns per month

Engine mounting insulator bolt not getting pre fit/ getting slip during tightening on chassis at Engine drop stage

Cross Functional Team

No.	Description	Name	Department	Mobile Number
1.	**Team Leader**	**Sandip Borse**	**S/X TCF PRODUCTION**	**9753139602**
2.	Mentor	Taj Shaikh.	S/X TCF PRODUCTION	8378991344
3.	Member 1	Sachin Bhamre	S/X TCF MQA	9158960111
4.	Member 2	Satish Pagar	S/X TCF SQA	9822209653
5.	Member 3	Darshak Thakkar	S/X TCF SQA	7720074994
6.	Member 4	Arvind P. Mistry	CDMM	9922957031
7.	Member 5	Ajit Chaudhari	S/X TCF PRODUCTION	7588616388

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice (reasons)

Problem Criticality (Severity)

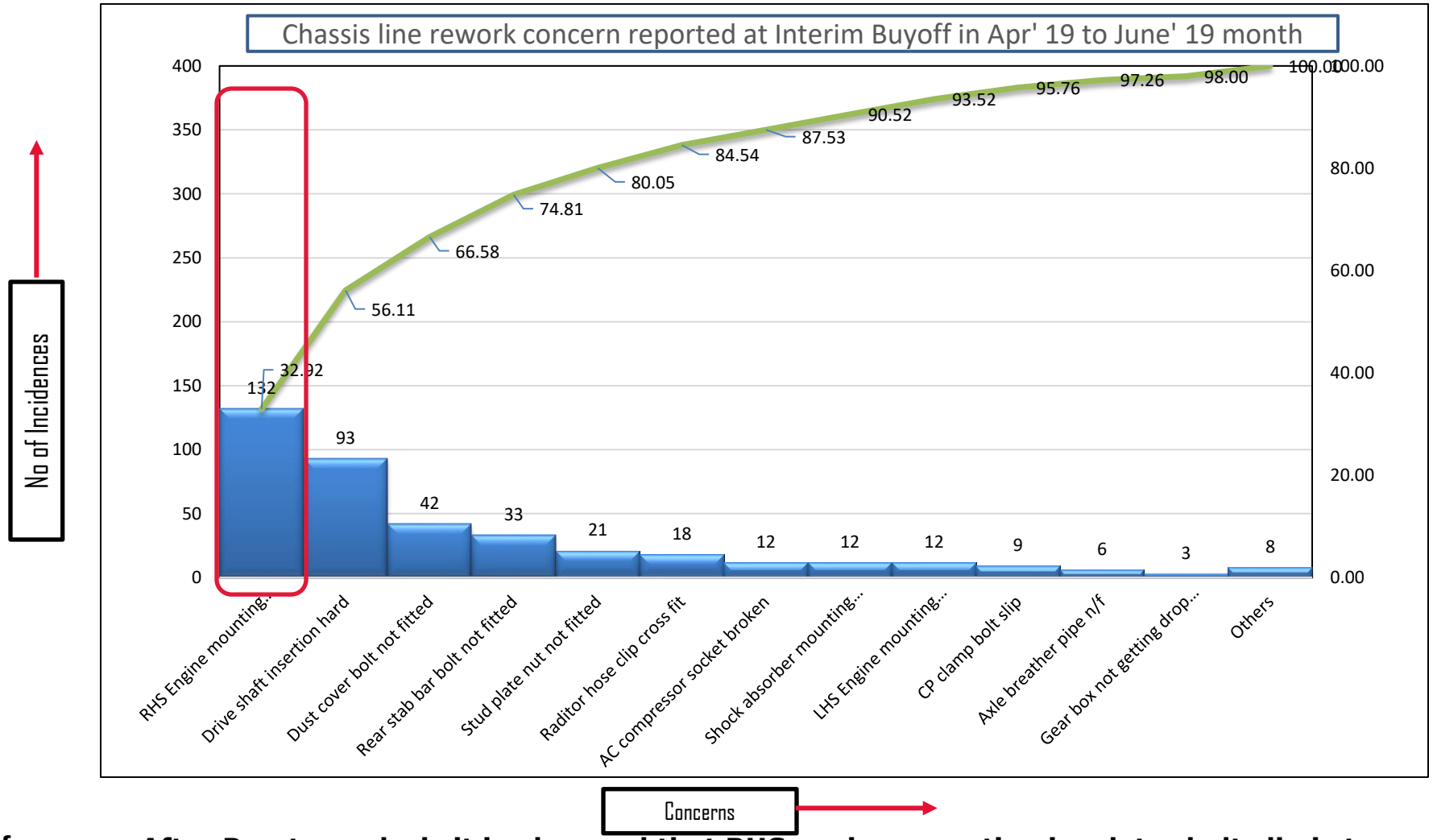
- **Importance & Severity of Concern :-**

- **Engine mounting insulator bolt not fit/ slip/ broken is V1+ concern.**
- **Major contributor in productivity loss of chassis line**
- **Major contributor in FRC of chassis line**
- **Avg. 44 cases reported in month**
- **Skill operator required for rework.**
- **Rework activity is critical in online as well as not possible in offline activity**
- **Repair cost is very higher 38206.5/- RS per year.**

Rework Cost Calculations :-

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	30
Effective working time	Min	465
Operator required /day	nos	0.06
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	127.35
Avg no. of working days	days	25
Operator cost /month	Rs	3183.9
Operator Cost/year	Rs	38206.5

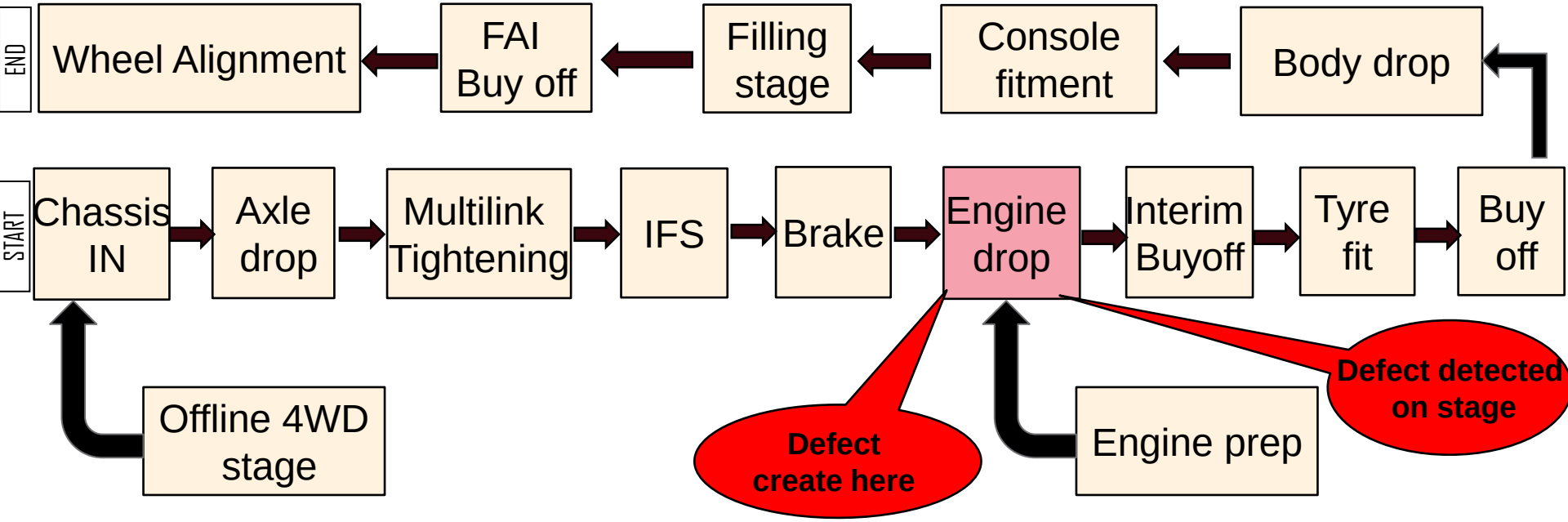
2. Justify the choice (reasons)



Inference – After Pareto analysis it is observed that RHS engine mounting insulator bolt slip is top concern from chassis line.

3. Understand the current situation

Observation of process



3. Understand the current situation

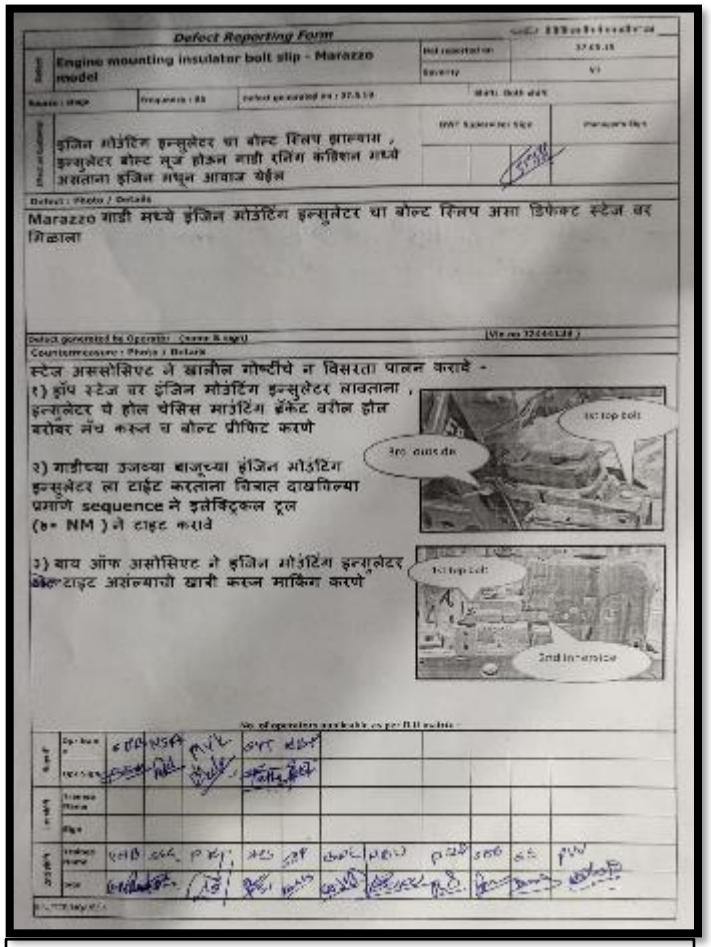
Observation of process flow

DEFECT CREATE HERE

IFS stage → Brake stage → Propeller shaft fitment → Engine Drop → Oil filing stage → Tyre Fitment → Power steering pipe fitment → Chassis buyoff

3.a Immediate Containment Action

- DRF & OPL given to operators for Concern detection at stage & buyoff.



Defect Reporting Form

Engine mounting insulator bolt slip - Marazzo model

इंजिन मौंटिंग इन्सुलेटर चा बोल्ट स्लिप झाल्यास , इन्सुलेटर बोल्ट लूज होऊन गाडी रनिंग कंडिशन मध्ये असताना इंजिन मधून आवाज येईल

Marazzo गाडी मध्ये इंजिन मौंटिंग इन्सुलेटर चा बोल्ट स्लिप असा डिफेक्ट स्टेज वर मिळाला

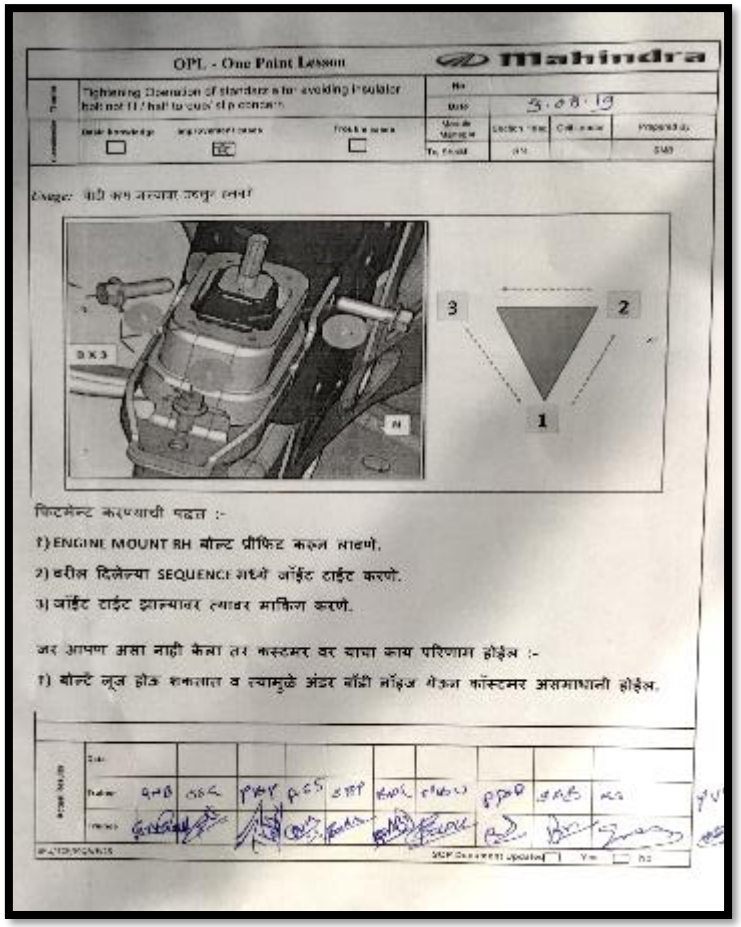
स्टेज असोसिएट ने खालील गोष्टींचे न विसरता पालन करावे -

१) ड्रॉप स्टेज वर इंजिन मौंटिंग इन्सुलेटर लावताना , इन्सुलेटर चे होल चेसिस माउंटिंग ब्रॅकेट वरील होल बरोबर मॅच करून च बोल्ट प्रीफिट करणे

२) गाडीच्या उजव्या बाजूच्या इंजिन मौंटिंग इन्सुलेटर ला टाईट करताना चित्रात दाखविल्या प्रमाणे sequence ने इलेक्ट्रिकल टूल (8- NM) ने टाइट करावे

३) बाय ऑफ असोसिएट ने इंजिन मौंटिंग इन्सुलेटर बोल्ट टाइट असल्याची खात्री करून मार्किंग करणे

Defect communication done through DRF to all stage associate. & Buy off associate.



OPL - One Point Lesson

Mahindra

Tightening Operation of standard nut for avoiding Insulator bolt not fit / half to cause of problem

Date: 3.08.19

Change: गाडी कम बनवावा उपयुक्त लेख

फिटमेन्ट करण्याची पद्धत :-

१) ENGINE MOUNT RH बोल्ट प्रीफिट करून घालणे.

२) दरील दिलेल्या SEQUENCE मध्ये बोल्ट टाईट करणे.

३) बोल्ट टाईट झाल्यावर त्यावर मार्किंग करणे.

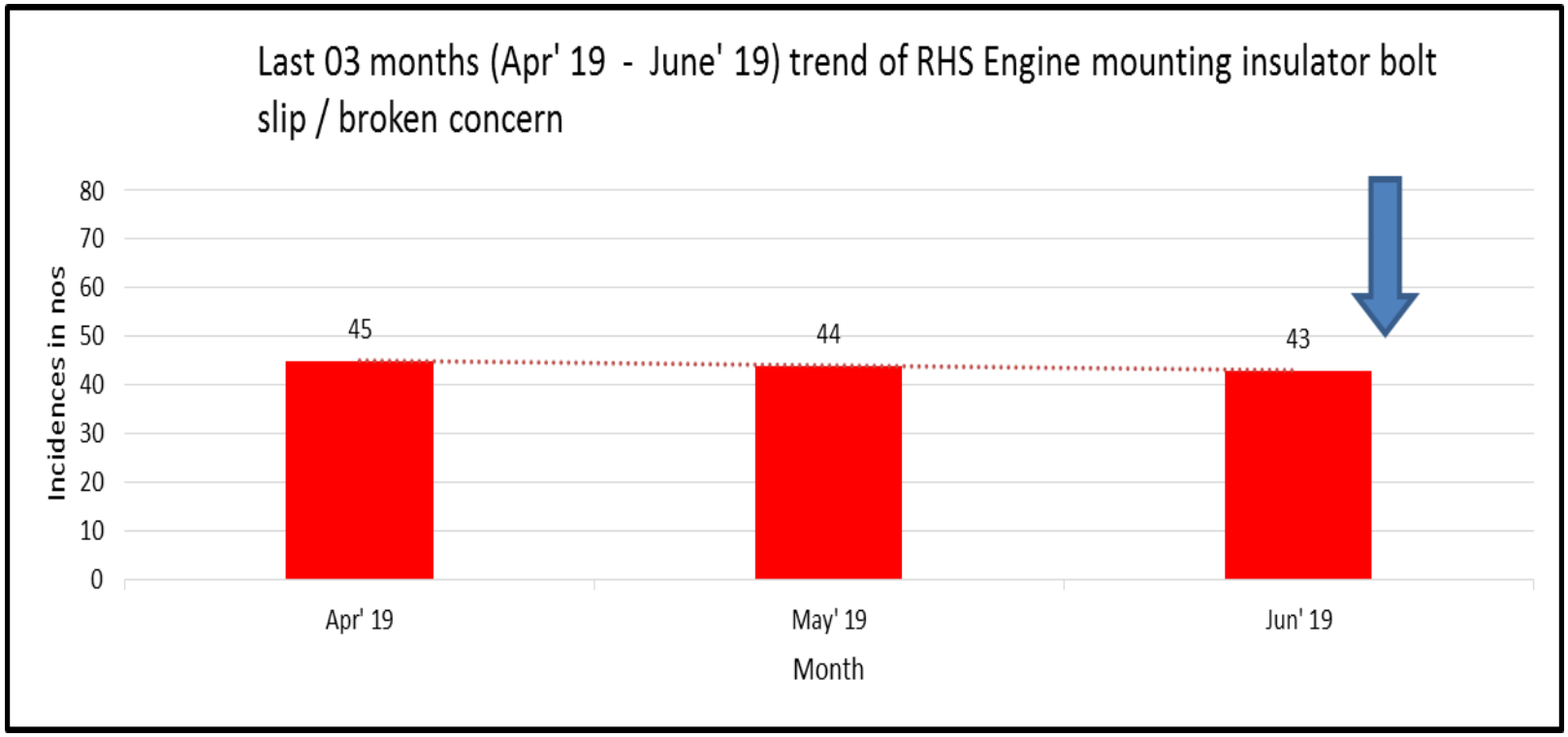
जर आपण असा नाही केला तर कस्टमर वर याचा काय परिणाम होईल :-

१) बोल्ट लूज होऊ शकतात व त्यामुळे अंडर बॉडी नॉइज येऊन कॉस्टमर असमाधानी होईल.

OPL given to all stage associate about engine mounting insulator bolt pre fitment and sequence tightening

3. Understand the current situation

BWT - 8 Chassis line rework concern reported at Interim Buyoff in Apr '-19 to June'-19 month



Inference – Average 44 concern per month reported buyoff ,
Avg. DPT of 3 months 132

3. Understand the current situation

L0 for Intercooler hose insertion hard concerns at interim buyoff stage in Marazzo Model.

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : Scorpio/Xylo TCF
Cell / BWT / Line: BWT 8

Requested By - MQA | Date - 15.05.19 | Source - Stage | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Engine mtg insulator bolt slip

FMEA Severity /Gravity : V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 2/3' | VIN / Aggre. No.- J2K44139 | Model : Marazzo

Date of Incidence/ Assy : 15.05.19 | Shift / Time : Both | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :
Rework done on stage

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - | BWT . :

Does a control already exist ? ☐ YES ☑ NO

Stage / Post Analysis -

Operator Name : Samadhan Padekar

Man(Human Action)

Is operator trained ?	☑ YES	☐ NO
Operator level (I.L.U)	Black I	
OR Dexterity level(1...4)	L3	
Is operator under training ?	☐ YES	☑ NO
Is operator temporary for Stage ?	☐ YES	☑ NO
Bad rework ?	☐ YES	☑ NO
Is the specified /Tool / Fixture facility used ?	☐ YES	☑ NO
Last calibration dates	☑ YES	☐ NO
Facility /Tool / Fixture in good condition ?	☑ YES	☐ NO
Is it a substitute facility ?	☐ YES	☑ NO
Are machine parameters known ? (Record in Analysis table)	☑ YES	☐ NO
Parameters are respected ?	☑ YES	☐ NO

Machine

Method(process)

Main steps are respected ?	☑ YES	☐ NO
Is there a related key point in SOP / SOS?	☑ YES	☐ NO
Are the reasons for key point known ?	☑ YES	☐ NO
Are operating reqmnts respected ?	☑ YES	☐ NO
Is the defect part of the Buy off / Checkman list ?	☑ YES	☐ NO
Was it detected by Inspector/ Buoff Opr ?	☑ YES	☐ NO
In only one shift ?	☐ YES	☑ NO
In only one line ?	☑ YES	☐ NO
The problem follows the vehicle ?	☐ YES	☑ NO
The problem follows the part ?	☐ YES	☑ NO
Is it a repeat problem ?	☑ YES	☐ NO
Is the suspected parameter measurable ? (Record in Analysis table)	☐ YES	☐ NO

Product Failure

Other

Is there a Control Plan ? (Record in Analysis table)	☑ YES	☐ NO
Is the packaging standard defined for the part ?	☑ YES	☐ NO
Is the packaging standard respected ?	☑ YES	☐ NO
Is there any alert mechanism for failure ?	☑ YES	☐ NO
Is the specified product used ?	☑ YES	☐ NO
Did it evolve / Change ?	☐ YES	☑ NO
Is the storage SOP respected ?	☐ YES	☑ NO
Is the shelf life / expiry date respected ?	☑ YES	☐ NO
Is the part as per FIFO?	☑ YES	☐ NO
	☑ YES	☐ NO

Material/ Consumable

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern

Analysis Conclusionon Gemba (Attach Sketch/ Photograph if necessary) -
Vehicle reported for - Engine mtg insulator bolt slip

Vehicle reworked by - Vehicle reworked by tapping and adding helicoil on mounting.

History - No

Process flow - Engine mounting insulator align on chassis mounting bracket, Pre fitment insulator bolt and tightening with 70 NM torque by electrical tool.

Detection - Engine not able to drop on chassis.

Traveler card - No history

Packaging - Good

Defect history - Nil

Failed part : RHS engine insualtor mounting hole not match with mounting bracket on chassis (stake tolerance issue between insulator mounting and

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark
1	Engine mounting insulator	70 +/- 5 NM	70 +/- 5 NM	

Is 4M Analysis Part 2 required ? ☑ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☐ NO | **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
1. Vehicle reworked by tapping and adding helicoil on mounting.			
1. Tightening of sequencing changed	SB / SS	15.05.19	Closed

2. Permanent Action	Resp	Date	Status
1. Part L1 required	Darshak Thakkar		open

Analysed by Name/s - S M Borse
Date - 15.05.19

Choice : ☑ WSS ☐ QC Story

- Inference from L0
Problem follows part
- Problem is a repeat problem
L1 analysis required.

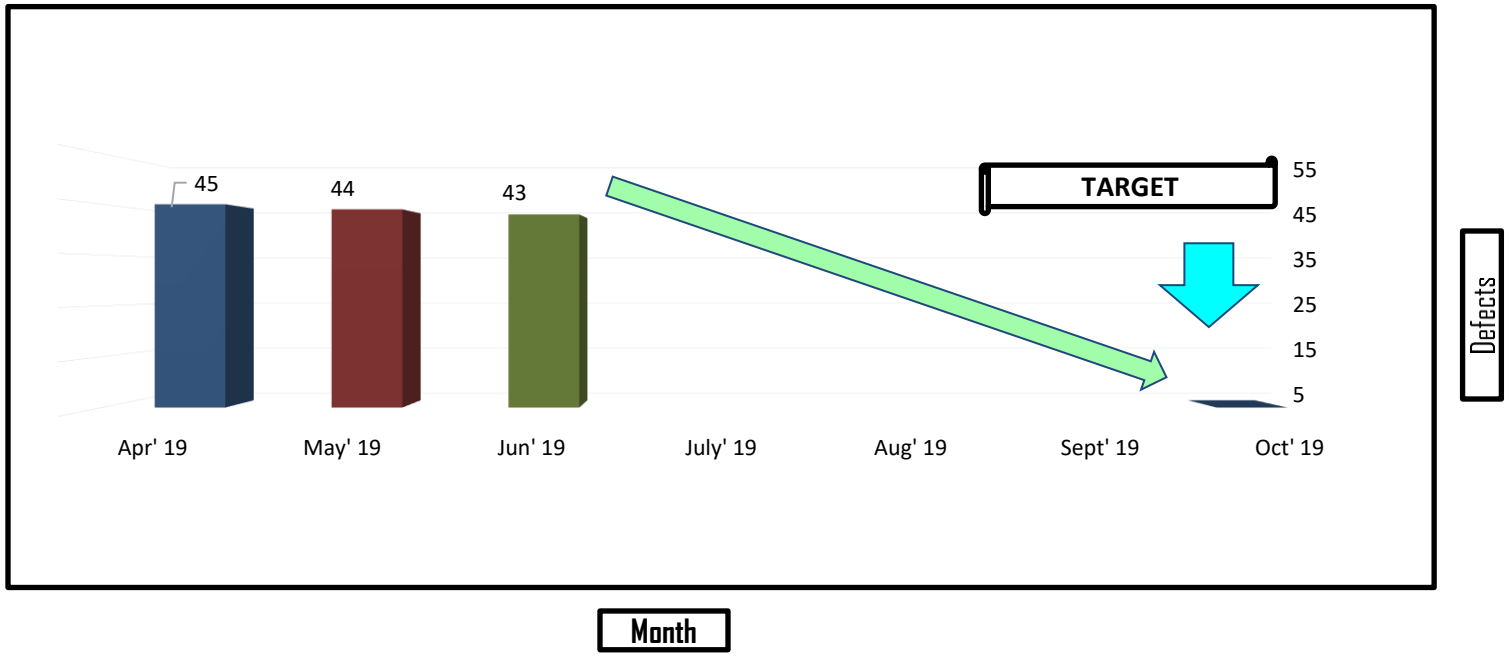
4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3/L4
Method	Intercooler hose white marking with engine turbo inlet notch
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part
Material	Material stored as per requirement.

No Quick Wins

4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Avg.44 vehicles / month reported for engine mounting insulator bolt slip concerns
Target:	To eliminate concern of engine mounting insulator bolt slip/ in Marazzo Model by Oct' 2019.



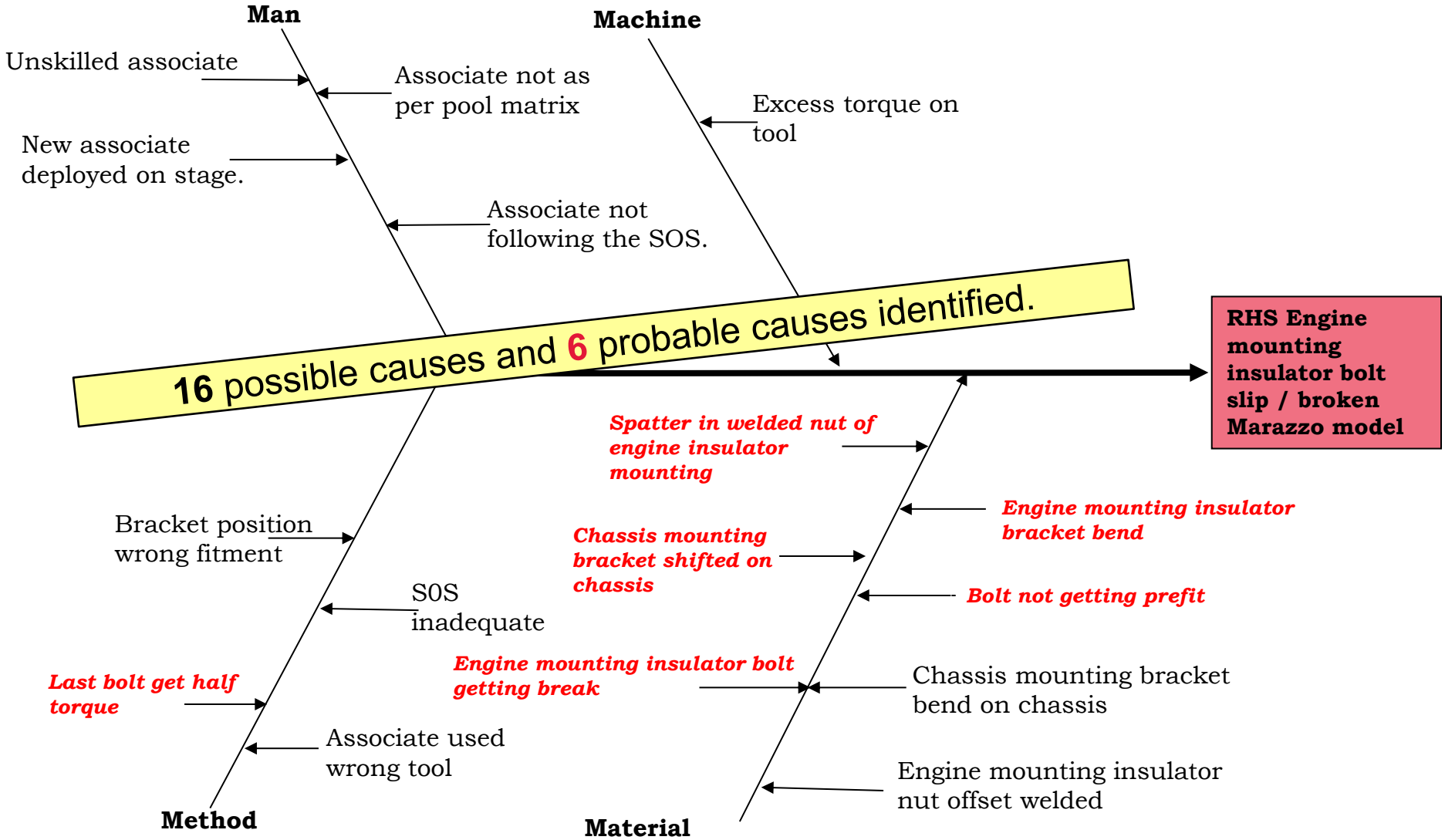
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	Unskilled associate
2	New associate deployed on stage
3	Associate not following the SOS.
4	Last bolt get half torque
5	associate not as per pool matrix
6	associate used wrong tool
7	Chassis mounting bracket bend on chassis
8	Engine mounting insulator bolt getting break
9	Bolt not getting prefit
10	Chassis mounting bracket shifted on chassis
11	Excess torque on tool
12	Engine mounting insulator bracket bend
13	Spatter in welded nut of engine insulator mounting
14	Engine mounting insulator nut offset welded
15	SOS In-adequate
16	Bracket position wrong fitment

5. Analysis

5.1 Identify possible causes: Fish Bone diagram.



5. Analysis

5.1 List of possible causes

Sr. no	Possible Causes	Sandip Borse	Taj shaikh	Sachin Bhamre	Satish Pagar	Abhyuday Shukla	K B Gawali	P V Vidhate	Total
1	Unskilled associate	1	1	3	3	3	3	3	**17**
2	New associate deployed on stage	1	1	1	1	1	1	1	**7**
3	Associate not following the SOS.	3	3	3	3	3	9	9	**33**
4	Last bolt get half torque	3	9	3	9	3	9	9	**45**
5	associate not as per pool matrix	1	1	1	1	1	1	1	**7**
6	associate used wrong tool	1	3	3	3	3	3	3	**19**
7	Chassis mounting bracket bend on chassis	1	1	3	3	3	3	3	**17**
8	Engine mounting insulator bolt getting break	1	3	9	9	9	3	9	**43**
9	Bolt not getting prefit	3	9	3	9	3	9	9	**45**
10	Chassis mounting bracket shifted on chassis	3	9	3	9	3	9	3	**39**
11	Excess torque on tool	1	1	1	3	1	1	3	**11**
12	Engine mounting insulator bracket bend	3	9	3	5	3	9	9	**41**
13	Spatter in welded nut of engine insulator mounting	3	9	9	3	3	9	3	**39**
14	Engine mounting insulator nut offset welded	1	1	3	3	3	3	3	**17**
15	SOS In-adequate	1	3	1	3	3	1	3	**15**
16	Bracket poition wrong fitment	3	1	3	1	3	1	3	**15**

Scale: 0=None, 1=Low, 3=Moderate, 9=Strong
Threshold Limit = 39

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes
1	Unskilled associate
2	New associate deployed on stage
3	Associate not following the SOS.
5	associate not as per pool matrix
6	associate used wrong tool
7	Chassis mounting bracket bend on chassis
11	Excess torque on tool
14	Engine mounting insulator nut offset welded
15	SOS In-adequate
16	Bracket poition wrong fitment

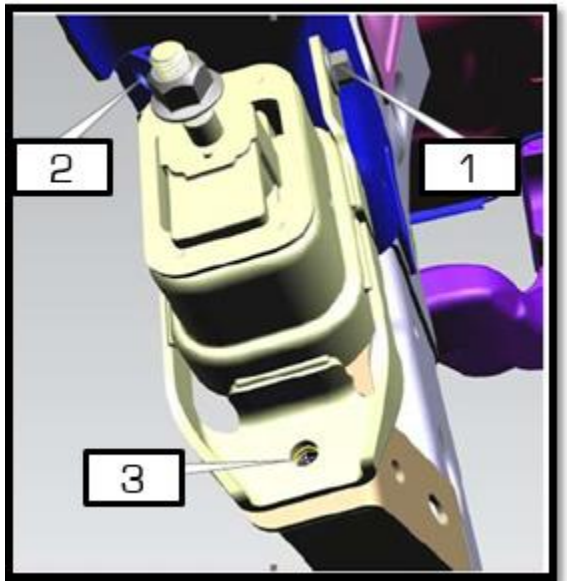
5.1 TEST OF HYPOTHESIS

Confirm the probable cause:-

Sr.No	Probable Cause
4	Last bolt get half torque
8	Engine mounting insulator bolt getting break
9	Bolt not getting prefit
10	Chassis mounting bracket shifted on chassis
12	Engine mounting insulator bracket bend
13	Spatter in welded nut of engine insulator mounting

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Last bolt get half torque	Visual Checking at Gemba	In 25 vehicles checked , 7 Vehicle found has tightening issue due to wrong sequence of tightening	Valid



Engine mtg. insulator bolt

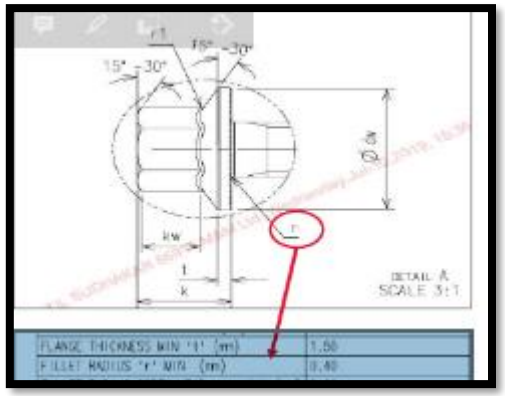
Probable cause			Testing Method		
Bolt not gets pre fit			Visual check at gemba		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	N.OK	19	OK
2	OK	11	N.OK	20	OK
3	OK	12	OK	21	OK
4	N.OK	13	OK	22	N.OK
5	OK	14	N.OK	23	OK
6	N.OK	15	OK	24	OK
7	OK	16	OK	25	N.OK
8	OK	17	OK		
9	OK	18	OK		

Inference – 25 Vehicles monitored , observed 7 vehicle tightening issue, due to wrong sequencing of tightening

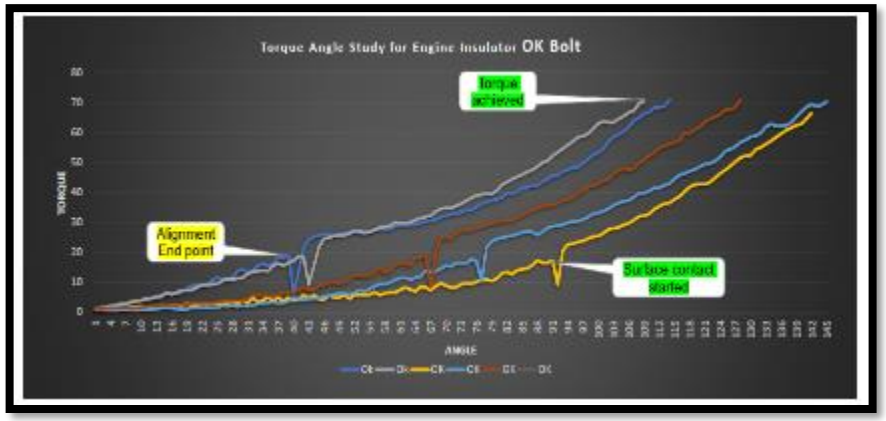
5.1 Test of hypothesis

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
8	Engine mounting insulator bolt getting break	Torque study on Hecta machine	Out of 25 vehicles of which bolts broken in 2 vehicles	**Valid**



Engine mounting insulator bolt getting break

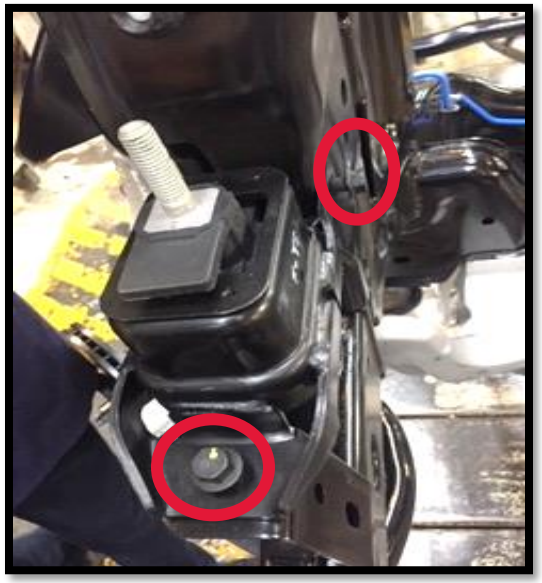


Torque study done for 25 vehicles out of which bolts broken in 2 vehicles

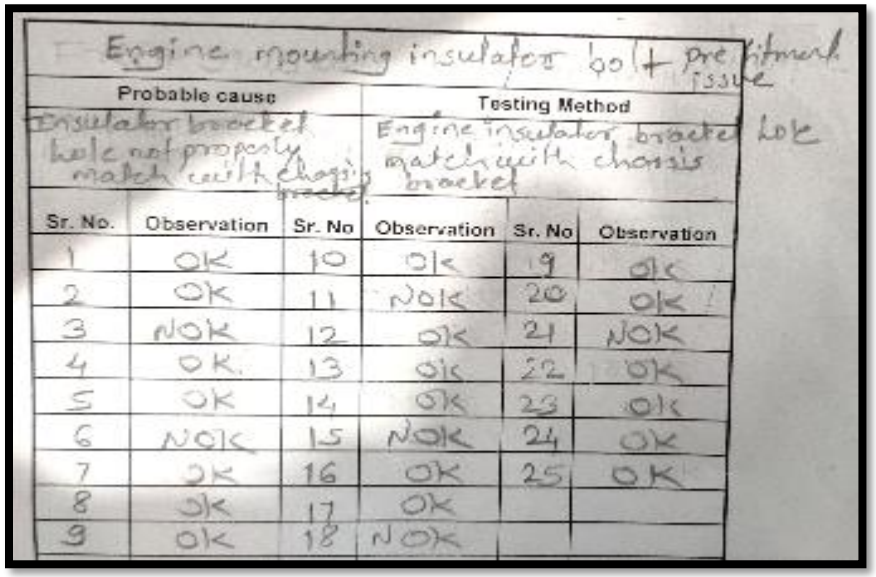
Inference – There is a point contact in NG condition, leads to increase in bolt preload

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
9	Bolt not getting pre fit	Physical check with Ok dimensional insulator bracket	Out of 25 Samples , found 6 engine mounting insulator hole not match with chassis bracket.	Valid



Checked with insulator



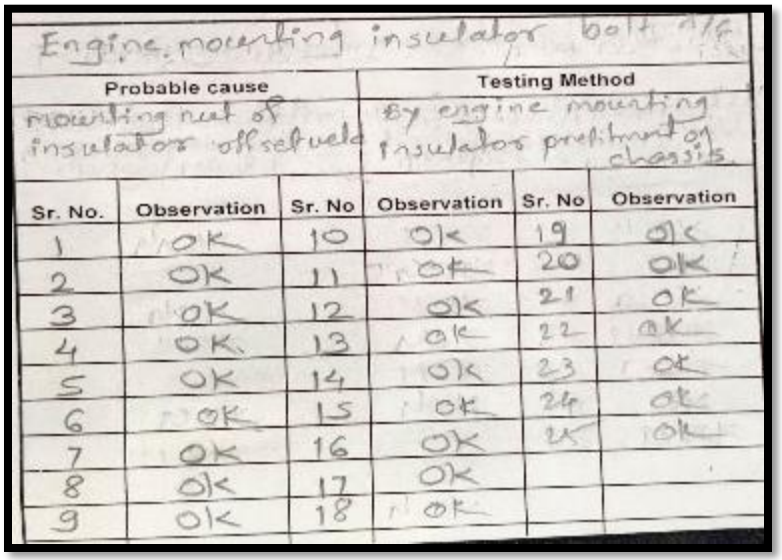
Engine mounting insulator bolt pre fitment issue

Probable cause			Testing Method		
Insulator bracket hole not properly match with chassis bracket			Engine insulator bracket hole match with chassis bracket		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	OK
2	OK	11	NOK	20	OK
3	NOK	12	OK	21	NOK
4	OK	13	OK	22	OK
5	OK	14	OK	23	OK
6	NOK	15	NOK	24	OK
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	NOK		

Inference – 06 Engine mounting insulator bracket hole not match with chassis bracket

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
10	Engine mounting insulator chassis bracket nut offset	Prefitment of calibrated engine mounting insulator on chassis	RHS engine mounting insulator weld nut offset check in 25 vehicles by pre fitment of bolt found all ok out of **25** vehicles	**Invalid**



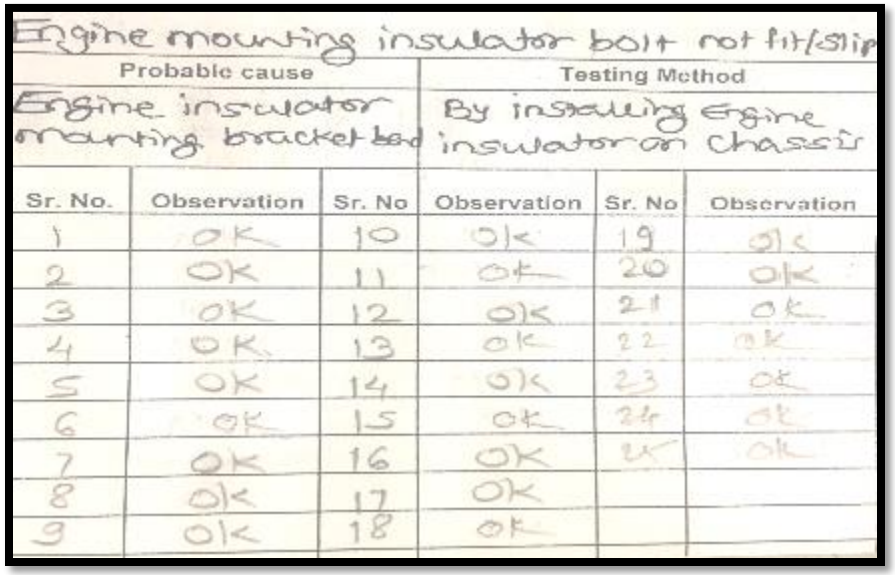
Engine mounting insulator bolt

Probable cause		Testing Method			
mounting nut of insulator offset weld		By engine mounting insulator prefitment on chassis			
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	OK
2	OK	11	OK	20	OK
3	OK	12	OK	21	OK
4	OK	13	OK	22	OK
5	OK	14	OK	23	OK
6	OK	15	OK	24	OK
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	OK		

Inference – In 25 vehicles, Engine mounting insulator chassis bracket nut offset found ok condition hence this is a invalid cause.

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
12	Engine insulator mounting bracket bend	Visually, Physically installing prechecked engine mount insulator on Chassis	25 chassis checked, engine insulator bracket found ok	**Invalid**



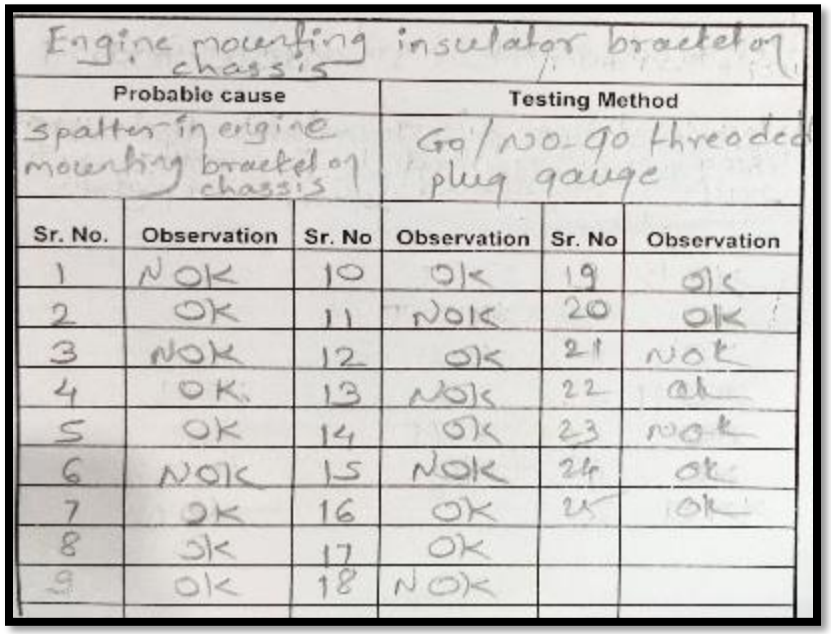
Engine mounting insulator bolt not fit/slip

Probable cause		Testing Method			
Engine insulator mounting bracket bend		By installing Engine insulator on Chassis			
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	OK
2	OK	11	OK	20	OK
3	OK	12	OK	21	OK
4	OK	13	OK	22	OK
5	OK	14	OK	23	OK
6	OK	15	OK	24	OK
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	OK		

Inference – 25 Chassis checked for insulator mounting bracket, found ok

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
13	Spatter in welded nut of engine insulator mounting	Inspect with threaded GO/ NOGO threaded plug gauge.	Found Spatter in **09** Vehicle during pre fitment out of **25** Vehicle .	Valid



Engine mounting insulator bracket on chassis

Probable cause			Testing Method		
spatter in engine mounting bracket on chassis			Go/NO-go threaded plug gauge		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	NOK	10	OK	19	OK
2	OK	11	NOK	20	OK
3	NOK	12	OK	21	NOK
4	OK	13	NOK	22	ok
5	OK	14	OK	23	nok
6	NOK	15	NOK	24	ok
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	NOK		

Inference – Spatter found in 09 Vehicle out of 25 vehicle, hence this is a Valid cause.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Last bolt get half torque	Visual observation	Hypothesis valid
2	Engine mounting insulator bolt getting break	Hecta	Hypothesis valid
3	Bolt not getting prefit	Physical check with Ok dimensional insulator bracket	Hypothesis valid
4	Spatter in welded nut of engine insulator mounting	By Go - No Go plug gauge	Hypothesis valid

4 Valid Cause

WHY - WHY Analysis

5.3 Identify the root causes: (Use Why – Why)

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
1	Last bolt get half torque	Excess stress on last bolt	Sequence on tightening not define	Not given in SOS of sequencing tightening	————————	Define the sequence
2	Engine mounting insulator bolt getting break	Excess friction	Engine mounting insulator bolt material not ok	exceeds tensile strength eventually breaking of bolt	greater root radius restricts the bolt flange	Correct the bolt
3	Bolt not getting prefit	Engine mounting insulator hole dimension variation	Bolt cross fit on bracket	Mounting bracket and chassis bracket hole not match properly	Stack tolerance issue	Change the hole size
4	spatter in welded nut of engine insulator mounting	Spatter accumulated on weld nut during bracket welding	Weld nut not cover to avoid accumulation of weld spatter	No such provision of masking of weld nut during welding activity	————	Masking to be provide

6. Implement the corrective measures

Possible Remedies Identified

Legends

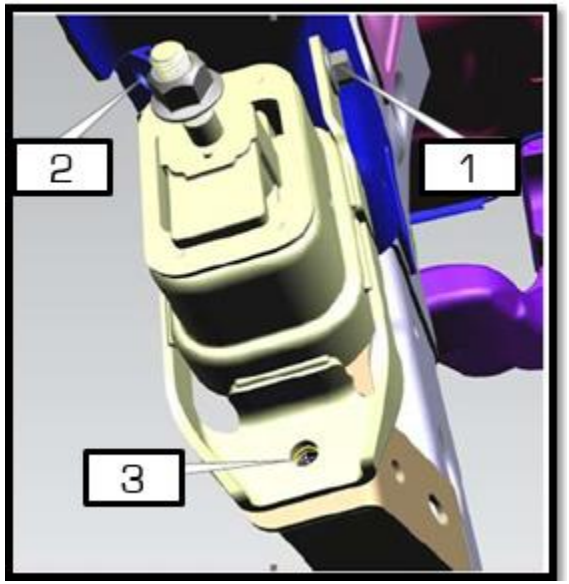
High ↑ Low ↓

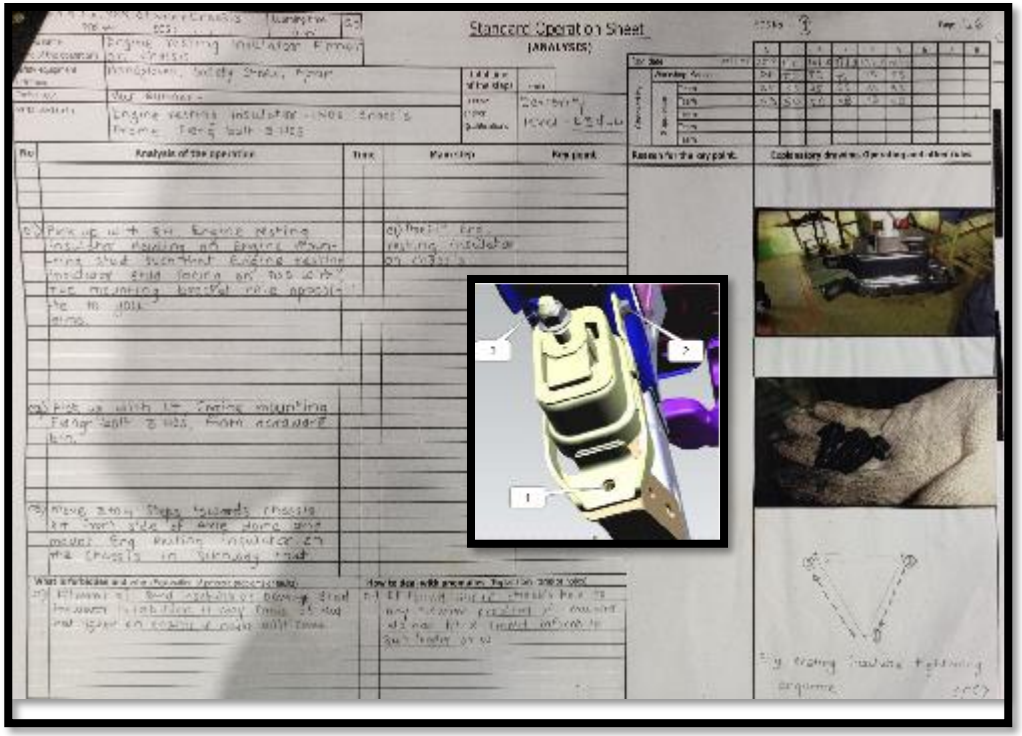
Sr. No.	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Sequence on tightening NOT OK in SOS	Define the sequence	Medium	Low	Yes	Low	**Accepted**
2	Engine mounting insulator bolt material not ok	Correct the bolt	Medium	High	No	Low	**Accepted**
3	Stack tolerance issue in engine insulator bracket	Change the hole size	Medium	High	Yes	Low	**Accepted**
4	Spatter in engine mounting insulator weld nut	Masking to be provide	Medium	low	Yes	Low	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Sequence on tightening NOT OK in SOS	Define the sequence	12.07.19	Sandip Borse	Closed





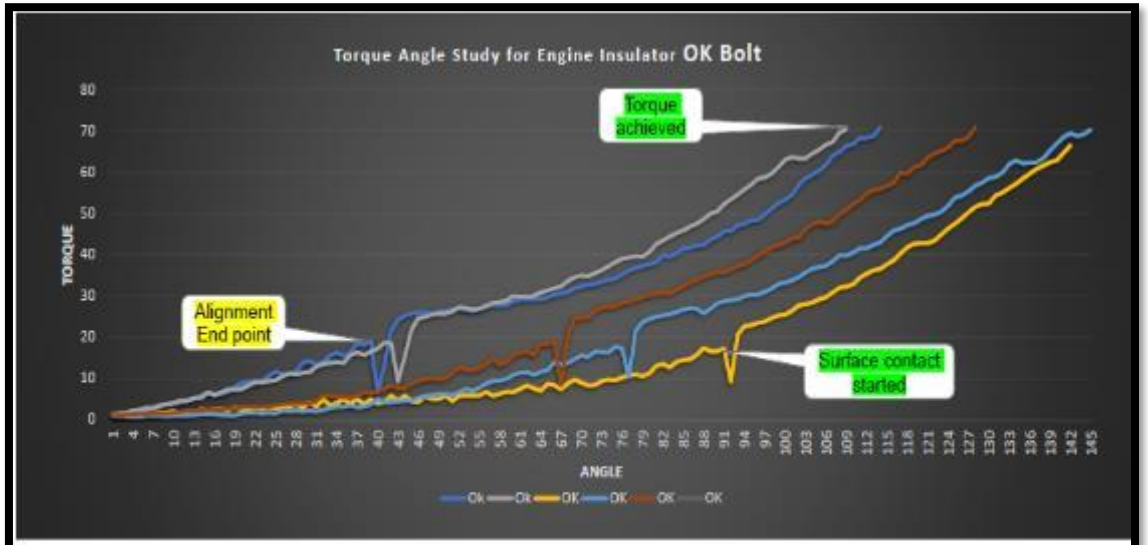
Define sequence of tightening

Updated in SOS

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Engine mounting insulator bolt material not ok	Bolt property to be correct	24th Aug 19	Shravan walte / Abhyuday	Closed



INFERNCE :

➢In Ok condition there is surface contact after alignment hence torque drop & gradual clamping by overcoming thread & underhead friction.

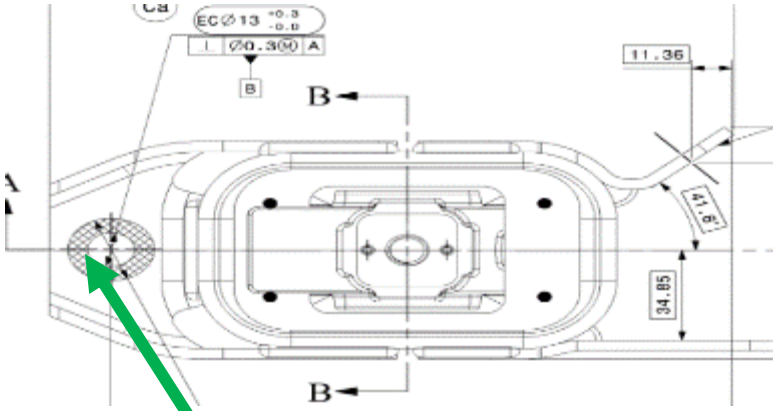
OK Condition with SKS made bolts

SKS made bolt started with root radius than 0.4mm

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Stack tolerance issue in engine insulator bracket	Hole size increased from 11mm to 13 mm	10th Sep'19	Shravan walte / Abhyuday	Closed

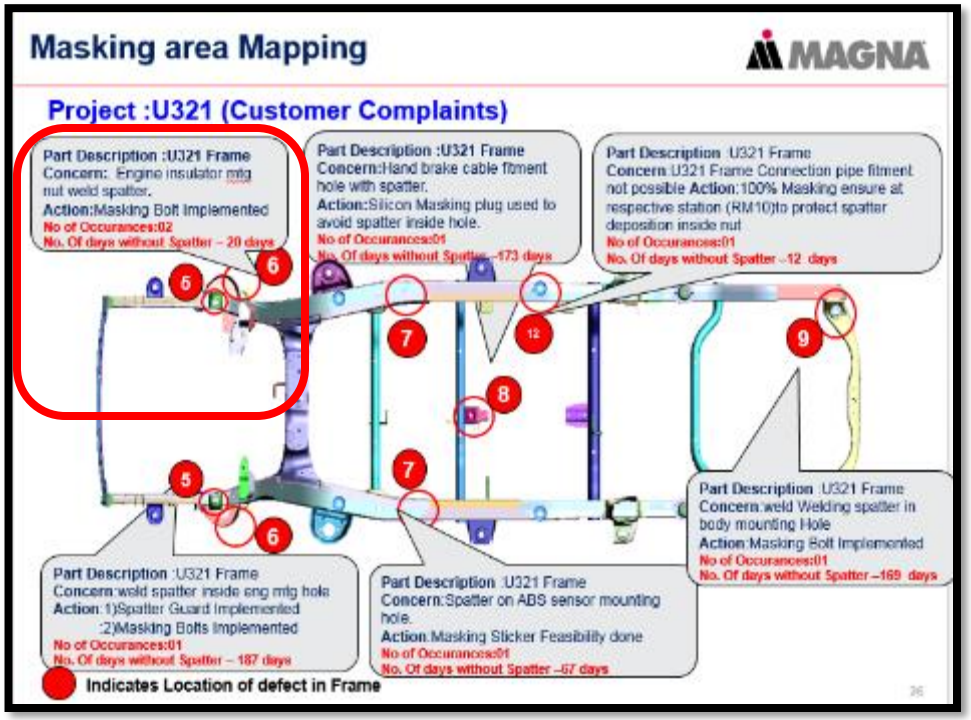


REV	COUNT	FIELD	CR-NO	CHANGES	DATE	NAME
A				RELEASE FOR PRODUCTION	22.12.17	SC/AK
B	4	B6,B12,G6 ,J5	20180522 02	GD&T POSITION VALUE INCREASED, DIMN. EC 11.00 +0.5/0 WAS DIMN.EC 11.00 +0.3/0 DIMN. SC 60.9±0.5 ,DIMN.SC 74.42±0.5 CRIMPING DIMN. REMOVED BASIC DIMN.5 ADDED, ASSY DIMN.72.4±1 ADDED.	30.05.18	T.S/AK
Ca	4	A2, G15,R2, R24	20190708010	MOUNTING HOLE DIA. CHANGED. DIM. Ø13 +0.3/-0 WAS Ø11 +0.3/-0, CHILD PART WEIGHT & TOTAL WEIGHT UPDATED. ** MARKED NOTE ADDED	10.09.19	T.S/VK
Cb	1	F15	20190708010	INCIDENTAL CYCLE ERROR CORRECTED	10.09.19	T.S/VK

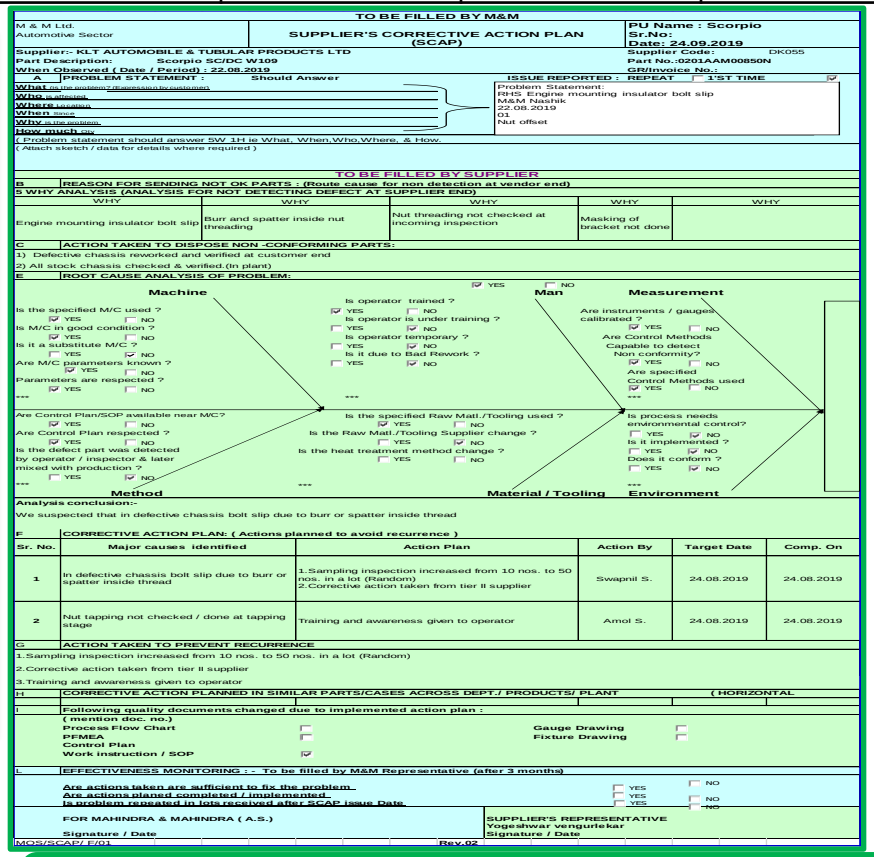
6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spatter in engine mounting insulator weld nut	Masking to be provide on nut to avoid spatter	15 July 19	Shravan walte / Abhyuday	Closed



Masking provided on engine insulator bracket nut



TO BE FILLED BY M&M

M & M Ltd. Automotive Sector | SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP) | PU Name : Scorpio, Sr.No:, Date: 24.09.2019

Supplier:- KLT AUTOMOBILE & TUBULAR PRODUCTS LTD — Supplier Code: DK055
Part Description: Scorpio SC/DC W109 — Part No.:0201AAM00850N
When Observed (Date / Period) : 22.08.2019 — GR/Invoice No.:

A PROBLEM STATEMENT : Should Answer — ISSUE REPORTED : REPEAT / 1'ST TIME ☑

What (is the problem? Description by customer)
Who (is affected)
Where (location)
When (when)
Why (is the problem)
How much (qty)

Problem Statement:
RHS Engine mounting insulator bolt slip
M&M Nashik
22.08.2019
01
Nut offset

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)

5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Engine mounting insulator bolt slip	Burr and spatter inside nut threading	Nut threading not checked at incoming inspection	Masking of bracket not done	

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:
1) Defective chassis reworked and verified at customer end
2) All stock chassis checked & verified.(In plant)

E ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine: Is the specified M/C used ? ☑ YES ☐ NO; Is M/C in good condition ? ☑ YES ☐ NO; Is it a substitute M/C ? ☐ YES ☑ NO; Are M/C parameters known ? ☑ YES ☐ NO; Parameters are respected ? ☑ YES ☐ NO

Man: Is operator trained ? ☑ YES ☐ NO; Is operator is under training ? ☐ YES ☑ NO; Is operator temporary ? ☐ YES ☑ NO; Is it due to Bad Rework ? ☐ YES ☑ NO

Measurement: Are instruments / gauges calibrated ? ☑ YES ☐ NO; Are Control Methods Capable to detect Non conformity? ☑ YES ☐ NO; Are specified Control Methods used ☑ YES ☐ NO

Method: Are Control Plan/SOP available near M/C? ☑ YES ☐ NO; Are Control Plan respected ? ☑ YES ☐ NO; Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO

Material / Tooling: Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO; Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO; Is the heat treatment method change ? ☐ YES ☐ NO

Environment: Is process needs environmental control? ☐ YES ☑ NO; Is it implemented ? ☐ YES ☑ NO; Does it conform ? ☐ YES ☑ NO

Analysis conclusion:-
We suspected that in defective chassis bolt slip due to burr or spatter inside thread

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	In defective chassis bolt slip due to burr or spatter inside thread	1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random) 2.Corrective action taken from tier II supplier	Swapnil S.	24.08.2019	24.08.2019
2	Nut tapping not checked / done at tapping stage	Training and awareness given to operator	Amol S.	24.08.2019	24.08.2019

G ACTION TAKEN TO PREVENT RECURRENCE
1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)
2.Corrective action taken from tier II supplier
3.Training and awareness given to operator

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL

I Following quality documents changed due to implemented action plan :
(mention doc. no.)
Process Flow Chart ☐
PFMEA ☐
Control Plan ☐
Work instruction / SOP ☑
Gauge Drawing ☐
Fixture Drawing ☐

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)
Are actions taken are sufficient to fix the problem ☐ YES ☐ NO
Are actions planed completed / implemented. ☐ YES
Is problem repeated in lots received after SCAP issue Date ☐ YES ☐ NO

FOR MAHINDRA & MAHINDRA (A.S.) — Signature / Date
SUPPLIER'S REPRESENTATIVE Yogeshwar vengurlekar — Signature / Date

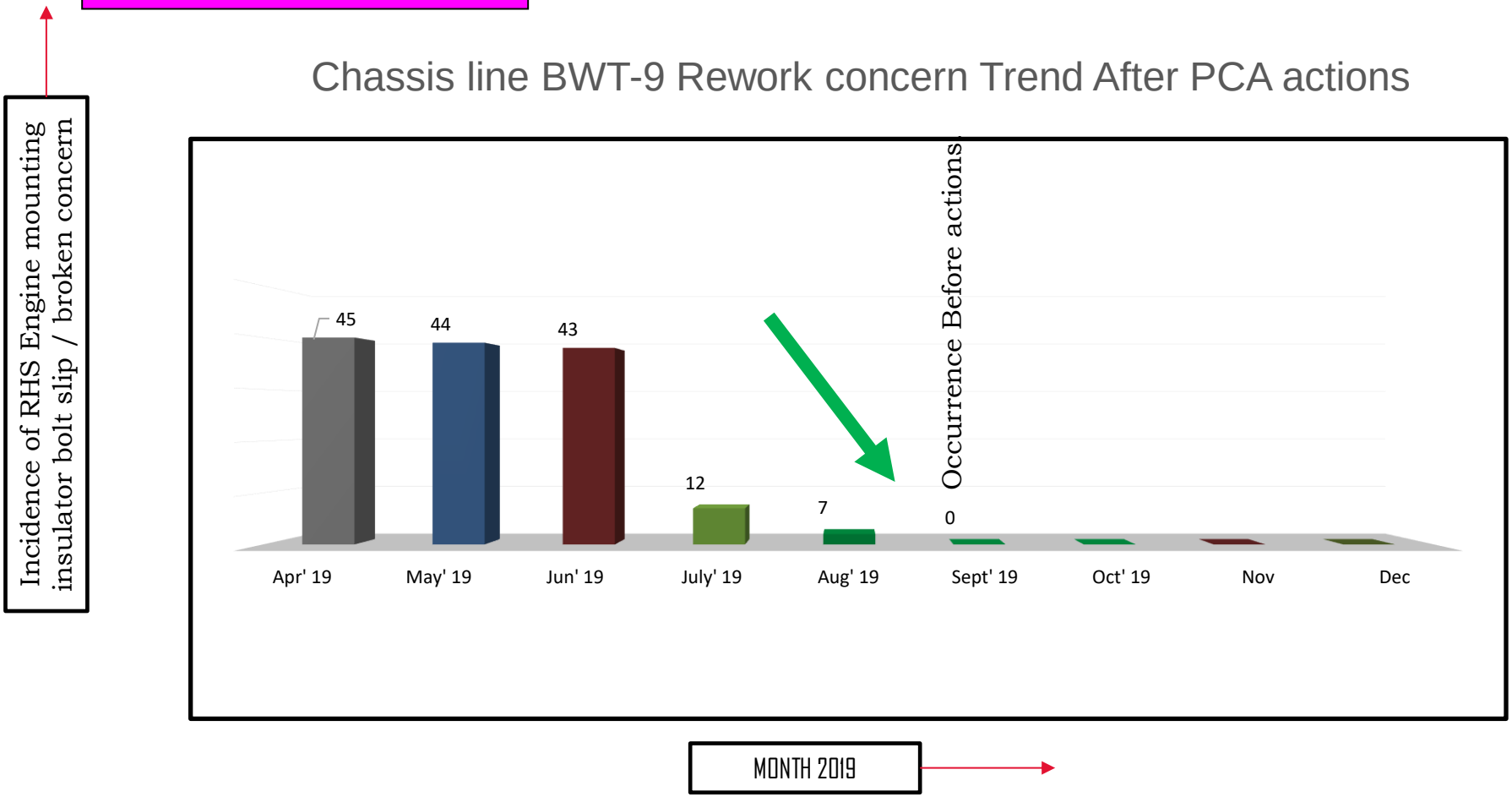
MOS/SCAP/ F/01 Rev.02

SCAPA of spatter in weld weld nut concern of insulator mounting

7. Check & confirm the results & effects

7.1 Verify Results:

Chassis line BWT-9 Rework concern Trend After PCA actions



Inference - After tacking PCA action, RHS Engine mounting insulator bolt slip / broken concern eliminated

7. Check & confirm the results & effects

7.3 Full Implementation Plan :

Action Plan	Target Dt	Resp.	Status
Engine mounting insulator tightening sequence to be change	12.07.19	Sandip Borse	Closed
Engine mounting insulator bolt property changed and make changed	24.08.19	Shailesh Kumar	Closed
Engine mounting insulator hole size to be increase from 11mm to 13 mm	10.08.19	Shailesh Kumar	Closed
SCAPA on spatter welding	15.07.19	Shailesh Kumar	Closed
PFMEA - RHS engine mounting insulator fitment	03.08.19	Sandip Borse	Closed
Process audit sheet - Sequence of tightening	03.08.19	Sandip Borse	Closed
OPL - given to stage associate	03.08.19	Sandip Borse	Closed
QA matrix	03.08.19	Sandip Borse	Closed

7.3 Tangible benefits:

- REWORK cost saving – Rs. 38206.5 / Year.
- Line loss eliminated
- **DPT of 132 is reduced to zero.**

Tue 12/24/2019 5:30 PM

DHANOKAR RAJESH

RE: cost estimation

To BORSE SANDIP

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	30
Effective working time	Min	465
Operator required /day	nos	0.06
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	127.35
Avg no. of working days	days	25
Operator cost /month	Rs	3183.9
Operator Cost/year	**Rs**	**38206.5**

7.3 Intangible benefits:

1. Operator fatigue reduced due to no rework for engine mounting insulator bolt broken/slip concern online/offline.
2. Customer satisfaction at stage.

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing

SN	Document	Actual date	Responsibility	Status
1	FTR	10.06.19	Shravan Valte	Closed
2	Supplier Audit check sheet	05.08.19	Darshak Thakkar	Closed
3	SOS.	03.08.19	Sandip Borse	Closed
4	PFMEA	03.08.19	Sandip Borse	Closed
5	QA Matrix Updation	03.08.19	Sandip Borse	Closed

8.2 Training

SN	Training Identified	Actual date	Responsibility	Status
6	One point lesson to Stage associate.	03.08.19	Sandip Borse	Closed

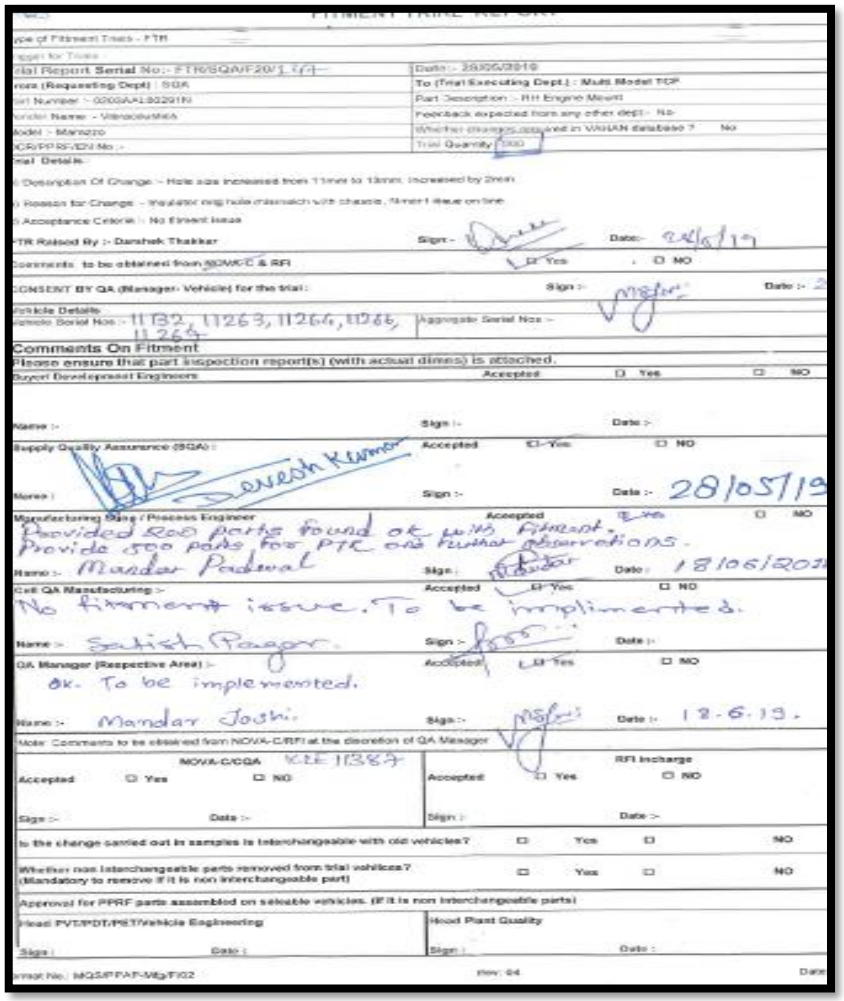
8.3 Additional Audit & check

SN	Training Identified	Actual date	Responsibility	Status
7	Process audit sheet	03.08.19	Sandip Borse	Closed
8	Daily monitoring supervisory Checksheet	03.08.19	Sandip Borse	Closed

8. Standardization

8.1 Updation of documents:

1. FTR



Type of Fitment Trials - FTR

Trial Report Serial No:- FTR/SQA/F20/1

From (Requesting Dept) : SQA

Part Number :- 0203AA130291N

Vendor Name - Vibracoustikka

Model - Marazzo

ECR/PP RF/EN No :-

Date :- 28/05/2019

To (Trial Executing Dept.) : Multi Model TCP

Part Description :- RH Engine Mount

Feedback expected from any other dept :- No

Whether changes updated in VAHAN database ? No

Trial Quantity 500

Trial Details

i) Description Of Change :- Hole size increased from 11mm to 13mm. Increased by 2mm

ii) Reason for Change :- Insulator mig hole mismatch with chassis, fitment issue on line

iii) Acceptance Criteria :- No Fitment issue

FTR Raised By :- Darshek Thakkar Sign :- Date :- 28/5/19

Comments to be obtained from NOVA-C & RFI ☑ Yes ☐ NO

CONSENT BY QA (Manager- Vehicle) for the trial : Sign :- Date :-

Vehicle Details

Vehicle Serial Nos :- 11132, 11263, 11264, 11266, 11267 Aggregate Serial Nos :-

Comments On Fitment

Please ensure that part inspection report(s) (with actual dimns) is attached.

Buyer Development Engineers Accepted ☐ Yes ☐ NO

Name :- Sign :- Date :-

Supply Quality Assurance (SQA) : Accepted ☑ Yes ☐ NO

Name : Sign :- Date :- 28/05/19

Manufacturing Shop / Process Engineer Accepted ☑ Yes ☐ NO

Provided 200 parts found ok with fitment. Provide 500 parts for PTR and further observations.

Name :- Mandar Padwal Sign : Date : 18/06/2019

Cell QA Manufacturing :- Accepted ☑ Yes ☐ NO

No fitment issue. To be implimented.

Name :- Satish Pawar Sign :- Date :-

QA Manager (Respective Area) :- Accepted ☑ Yes ☐ NO

ok. To be implemented.

Name :- Mandar Joshi. Sign :- Date :- 18-6-19.

Note: Comments to be obtained from NOVA-C/RFI at the discretion of QA Manager

NOVA-C/CQA KRE 11387 Accepted ☐ Yes ☐ NO Sign :- Date :-

RFI Incharge Accepted ☐ Yes ☐ NO Sign : Date :-

Is the change carried out in samples is interchangeable with old vehicles? ☐ Yes ☐ NO

Whether non interchangeable parts removed from trial vehicles? (Mandatory to remove if it is non interchangeable part) ☐ Yes ☐ NO

Approval for PPRF parts assembled on saleable vehicles. (If it is non interchangeable parts)

Head PVT/PDT/PET/Vehicle Engineering Sign : Date :

Head Plant Quality Sign : Date :

Format No.: MQS/PFAF-Mg/F02 Rev: 04 Date

REV	COUNT	FIELD	ECR-NO	CHANGES	DATE	NAME
A				RELEASE FOR PRODUCTION	22.12.17	SC/AK
B	4	B6,B12,G6 ,J5	20180522002	GD&T POSITION VALUE INCREASED, DIMN. EC 11.00 +0.5/0 WAS DIMN.EC 11.00 +0.3/0 DIMN. SC 60.9±0.5 ,DIMN.SC 74.42±0.5 CRIMPING DIMN. REMOVED BASIC DIMN.5 ADDED, ASSY DIMN.72.4±1 ADDED.	30.05.18	T.S/AK
Ca	4	A2, G15,R2, R24	20190708010	MOUNTING HOLE DIA. CHANGED. DIM. Ø13 +0.3/-0 WAS Ø11 +0.3/-0, CHILD PART WEIGHT & TOTAL WEIGHT UPDATED. ** MARKED NOTE ADDED	10.09.19	T.S/VK
Cb	1	F15	20190708010	INCIDENTAL CYCLE ERROR CORRECTED	10.09.19	T.S/VK

8. Standardization

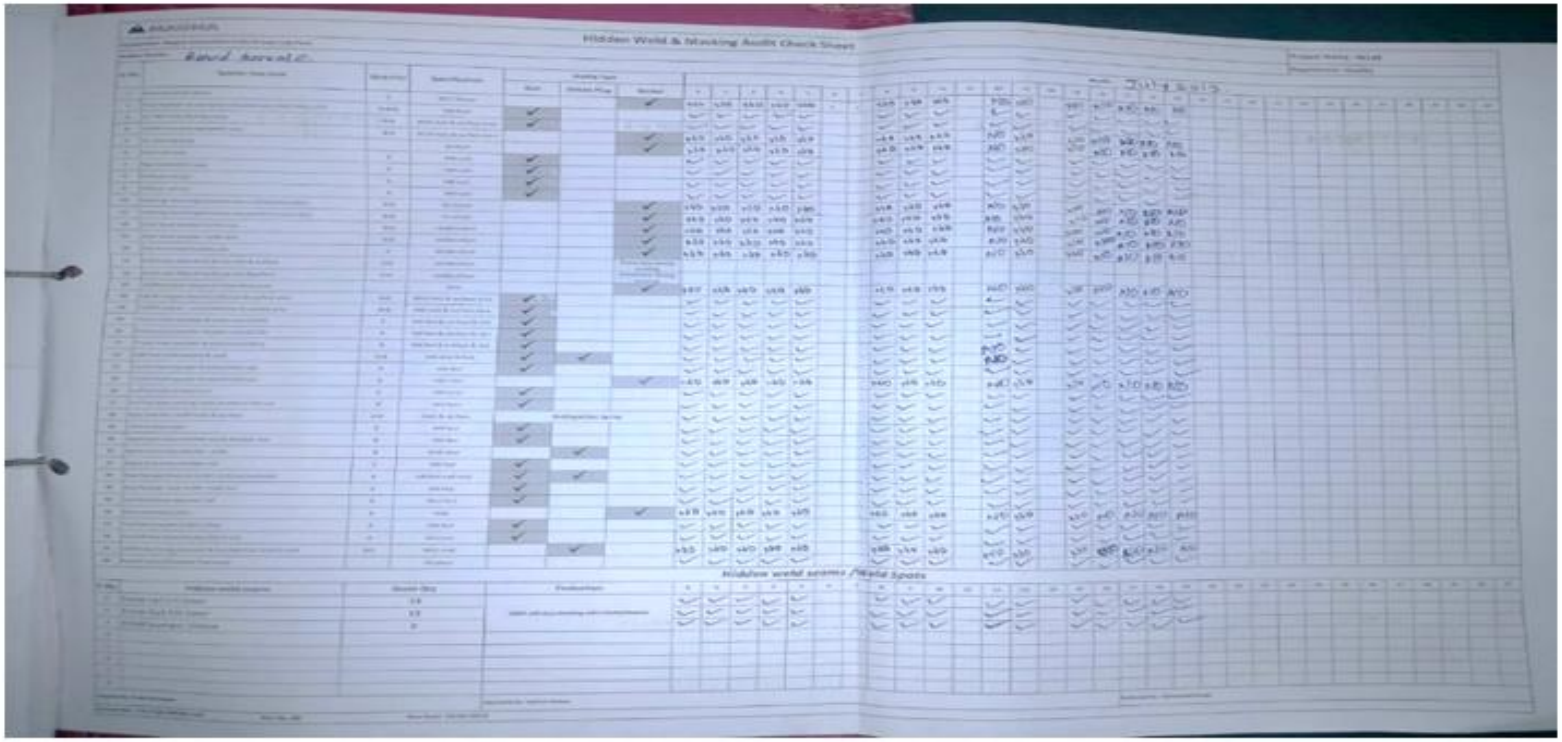
8.1 Updation of documents:

2. Supplier Audit check sheet

Action Plan for spatter reduction

MAGNA

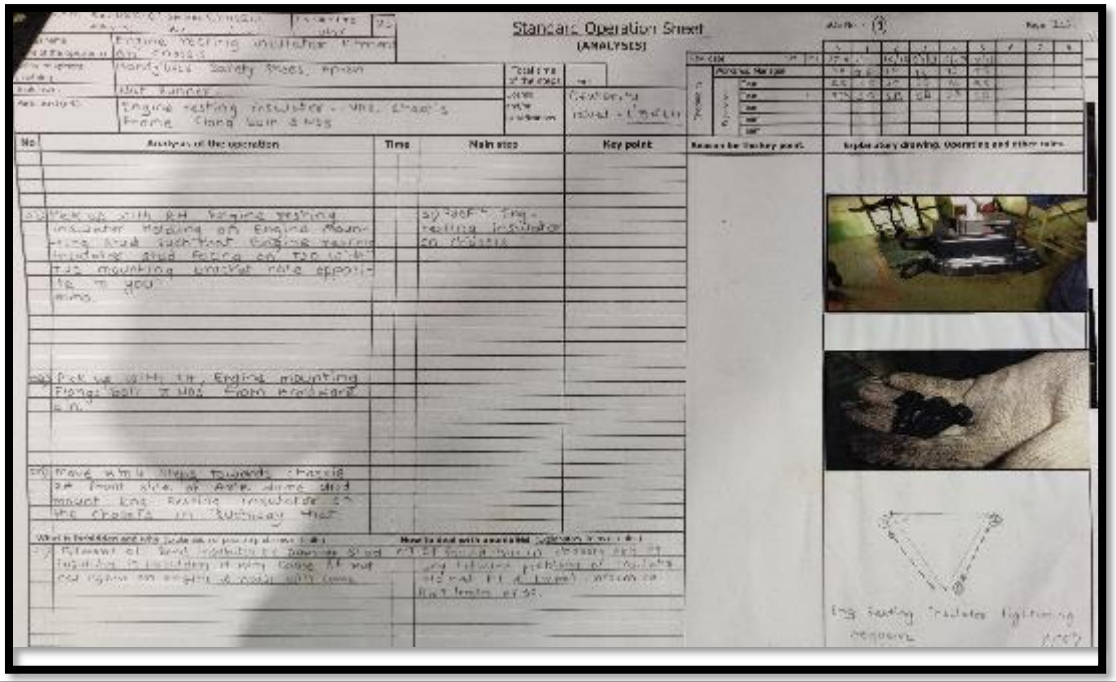
Masking Audit Check sheet

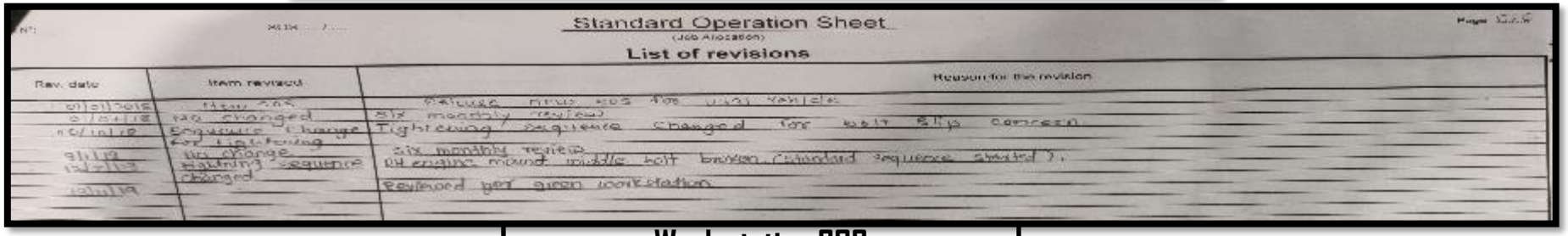


8. Standardization

8.1 Updation of documents:

3. SOS Updated :- RHS Engine mounting insulator fitment





Work station SOS

8. Standardization

8.1 Updation of documents:

4. FMEA- RHS Engine mounting insulator fitment

Plant : Nasik	Model : Marazzo	Mahindra & Mahindra Ltd.			Cross Functional Team	
Part No/s.:	Part Name/ Process: RH engine mount				Nihkil / Sor	Hari Nagar
Latest Change Letter: NA	Optn No. : 0203002	Process Potential Failure Mode & Effect Analysis.			Vikrant Ghule	Vikrant Morade
PU/ Shop : Scorpio/Xylo TCF	Optn Desptn: RH Engine mount insulator fitment	Reviewed by : Nikhil Hatedkar	Rev. No : 7	Doc No.: MQSODSU321TCF0203002		
Module/ Cell : TCF- Chassis	M/C Name & No.:	Approved by : Taj Shaikh	Rev. Date : 10.09.19	FMEA Date (Orig) : 15.05.2018		

Process Step / Function	'Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
RHS engine mount insulator fitment to chassis dome	Engine insulator loose	Bolt cross / slip	Mfg. & Assy : Under body Squeak and rattle Eng User - Engine vibration and underbody noise	9	cc	Associate did not prefit the bolt	Training and awareness given to all stage associates	3	Self check Marking on bolt head. And mark on bolt head at BUY OFF	7	189							
		Bolt loose	Mfg. & Assy : Under body Squeak and rattle Eng User - Engine vibration and underbody noise	9		Associate forget to tightening of mounting bolt	Training and awareness given to all stage associates	3	Self check Marking on bolt head. And mark on bolt head at BUY OFF	4	108	Poka-yoke tightening bolt	SS/AC/ MP	Closed	9	2	4	72
	Tightening of engine insulator to frame	tightening sequence not follow	Mfg. & Assy : Under body Squeak and rattle Eng User - Engine vibration and underbody noise	9		Associate tighten bolt without sequence	Training and awareness given to all stage associates	3	Self check	7	189							
		Torque over torque / under torque	Mfg. & Assy : Under body Squeak and rattle Eng User - Engine vibration and underbody noise	9		tool torque setting not ok	Tool calibration done by MQA	3	Self check Marking on bolt head. And mark on bolt head at BUY OFF	7	189							

Rev No.:	Rev Date	Revision Summary	Reason for Revision
1	15/05/2018	New PFMEA for U321 project	New release
2	16.11.18	No Change	Six monthly review
3	12.2.19	No Change	Six monthly review
4	12.07.19	Bolt tightning sequence change	Bolt slip/half torque
4	21.7.19	PCRN/A003/STCF/F20/13790	Reviewd for PCRN
5	5.8.19	No Change	Six monthly review
6	26.09.19	Bolt make changed from STL to SKS	STL make bolt broken or slip
7	10.09.19	Mounting hole oblong from 10 to 13 mm	Engine mounting insulator bolt pre fitment issue

8. Standardization

8.1 Updation of documents:

5. QA Matrix – RHS Engine mounting insulator fitment

QA MATRIX

PU NAME	STCF
DEPT NAME	
SECTION NAME	CHASSIS

INITIALIZATION DATE				
SIGNATURES				
Dept Head				
Module Manager				
Section Manager	1	2	3	4

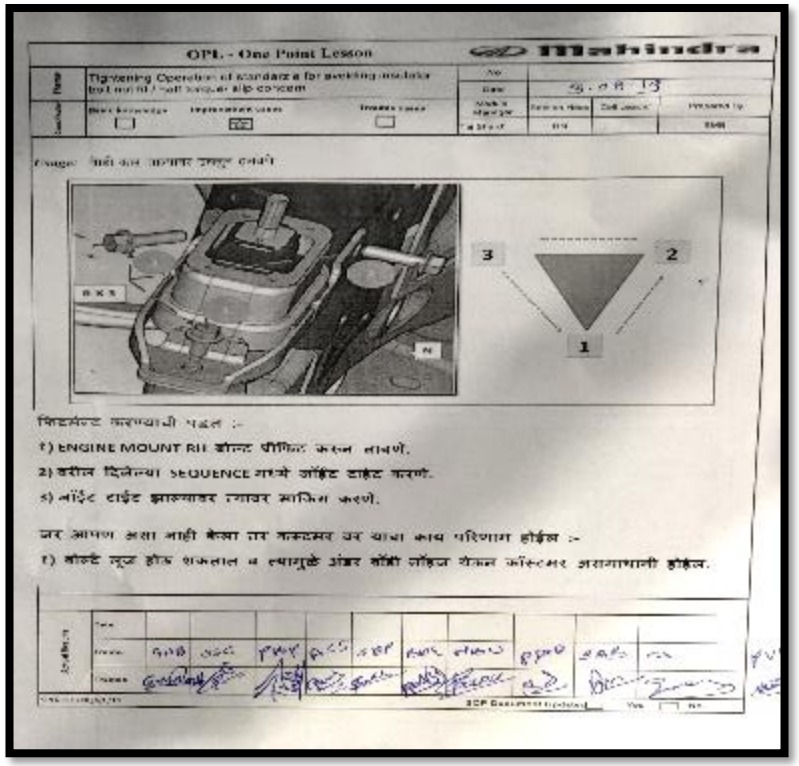
Global Quality Guarantee level.	
Section	Chassis
Plant	Nasik
Revision date	10.9.19

Workstation No.	Process description (POS)	Date	Failure modes	Severity	Prev. occurence	Potential decfect	Inspection process – Section: Axle	IFS	Inversion	Bundy tube	Propeller	Engine In	CSC	Engine prep	Engine drop	Interim buyo	Vin punchin	Oil	Tyre	Buyoff	Outside section: Air Cleaner	FAI Buyoff	CAI Underb	S&Q	WRAP	Smoke Test	Guarantee value: Workstation	Section	Outside sec	PLANT	Faults recorded: Section	Outside sec	CUSTOMER	Quality guarantee level: WORKSTAT	Section	Outside sec	PLANT	Comments
																															1M	1M	3M					
BWT 8	Engine mounting insualtor RH	18.09.18	Part not fitted/ Bolt half torue/ Bolt missing/ Bolt loose	C C	V 1 5										▽ 5					□ 1				□ 1			5	6	1	7	0	0	0	NOK	OK	NOK	OK	RH engine mount one bolt not visible , Out of three. 1. Electrical tool scanning base cycle available on stage.

8. Standardisation

8.2 Training:

Action Plan	Target Dt	Resp.	Status
6. OPL given to stage associate for RHS Engine mounting insulator fitment	3rd Aug 2019	Sandip Borse	Closed



OPL given to stage associate

8. Standardization

8.1 Updation of documents:

7. Process audit sheet modified for RHS Engine mounting insulator fitment.

PROCESS AUDIT SHEET BWT 8 & 9

BWT No.	Stage	Operation/Part	What to check	DATE							
8	ENGINE DROP	Roll restrictor bolt	Washer present on roll restrictor bolt.								
8		Propeller shaft	Ensure battery tool for 120HP bolt pretight and retorquing with torque wrench.								
8		Body mount	Tightning with scanning only & loose fitment all 7 bolts								
8		Engine Mounting Insulator	Ensure sequential tightning								
8		Drive shaft	No damage of rubber boot/uses of sleeve for smooth insertion.								
8		Engine Drop	LH side all 5 bolts tightning and match marking and ensure gun stop after all bolt tightning.								
8		LCA to UCA fitment	Presence of plain washer in Knuckle joint and UCA joint.								
8		IFS Tightning	Sequential tightning 50-90-50 Nm.								
8		Hub nut	Tightning 280 Nm and notching at buyoff.								
8		Shock absorber	Shock absorber bolt tightning and top cover fitment.								
9	TYRE & OIL	ABS sensor fitment	Bolt prefitment and no damage of bracket on stub axle								
9		Transaxle oil	Presence of plain aluminium washer in filler plug and pokayoke working								
9		Fuel line bracket	Orientation of fuel line bracket.								
9		Wheel nut fitment	As per model & retorquing ensure incase of not ok.								
9	ADDITIONAL	Brake leak test	Ensure complete cycle for checking & ensure no fouling or RH brake hose with drive shaft.								
		Oil & Urea filling	Ensure all pokayoke in working.								
		Buyoff	Ensure all check points are adhere.								

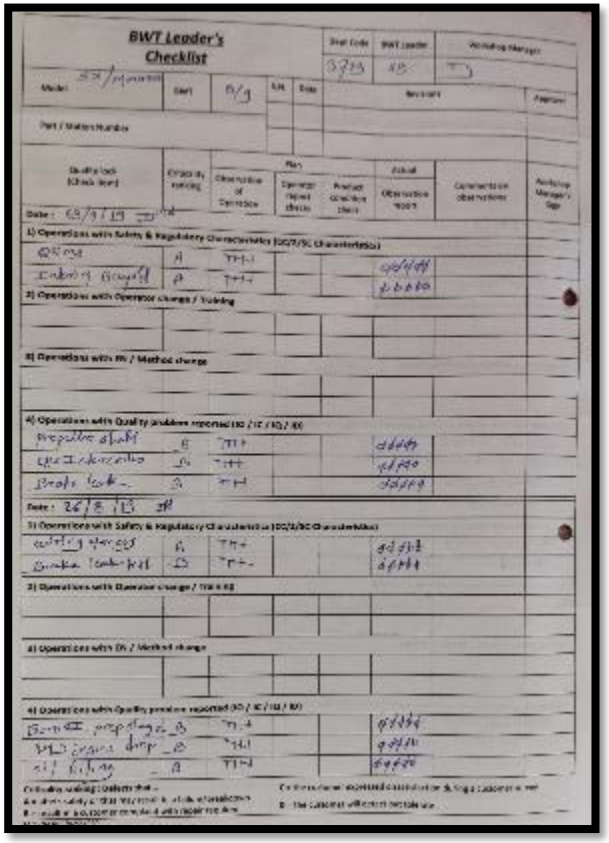
Engine mounting insulator tightening sequence checking point added in BWT Leader check sheet

8. Standardisation

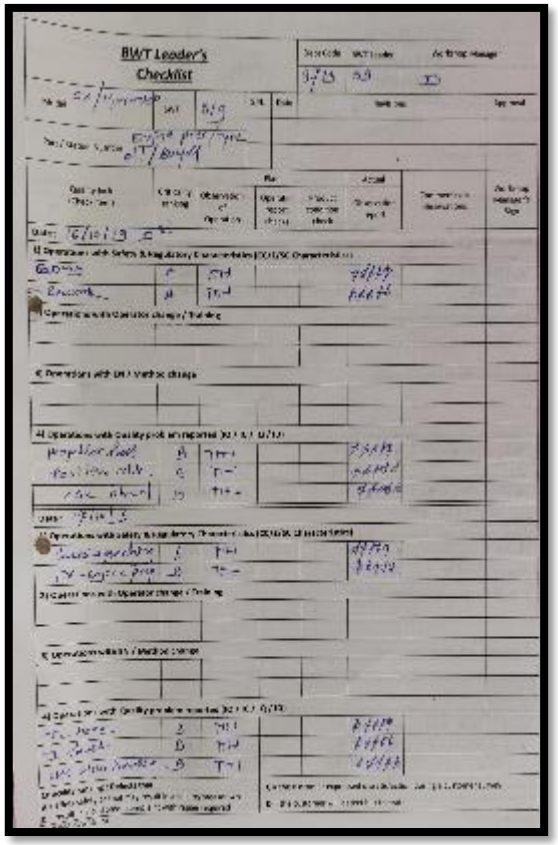
8.3 Additional Audit / Check

7. Point taken in daily supervisory check sheet –till NOV’18

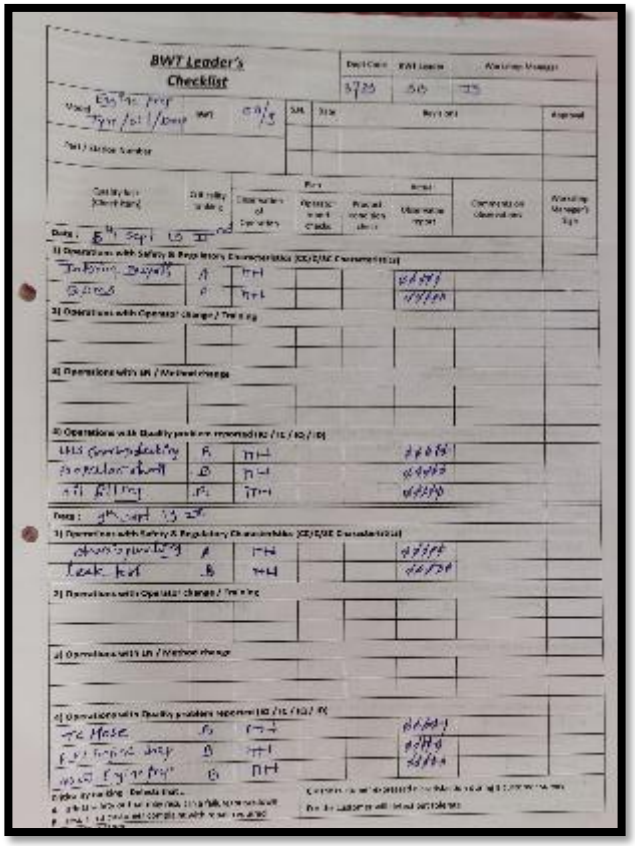
AUG 19



SEPT 18



OCT 18



9. Summarise & Plan future actions

9.1 Conclusion:

Engine mounting insulator concern in Marazzo model resolved by following actions.–

1. Bolt property changed and bolt Make change for elimination of bolt break concern.
2. Engine mounting insulator top mounting hole diameter changed for avoiding cross tightening or bolt thread rubbing
3. Engine mounting insulator tightening sequence change for elimination of rework and downtime on line

9.3 Horizontal Deployment:

1. W109 Vehicle at Nasik Plant - Chassis Line.

9.3 Lessons Learnt:

THINGS GONE WRONG: Variation in Material hardness and dimension variation in Insulator bracket hole get to difficult on fitment , no key point in SOS for proper sequencing of tightening, created for incomplete tightening / bolt broken.

THINGS GONE RIGHT: CFT worked in proper manner & resolved the concern RHS Engine mounting insulator bolt slip / broken in Marazzo model., Manual error eliminated.

Thank you……!